

Statement. There are 23 distinct real projective points and 23 distinct real projective lines, with precisely four selected incidences at every point and every line. The coordinates below lie in $\mathbb{Q}$.

23 POINTS P1,...,P23

P1 (1,0,0)
P2 (0,1,1)
P3 (0,1,-1)
P4 (0,2,3)
P5 (0,2,-3)
P6 (1,0,1)
P7 (1,0,-1)
P8 (3,-2,-2)
P9 (3,2,2)
P10 (3,-2,-2)
P11 (3,2,-2)
P12 (2,2,3)
P13 (2,-2,-3)
P14 (2,2,-3)
P15 (2,-2,3)
P16 (6,2,5)
P17 (6,-2,-5)
P18 (6,2,-5)
P19 (6,-2,5)
P20 (6,2,-1)
P21 (6,-2,1)
P22 (6,2,1)
P23 (6,-2,-1)

23 LINES L1,...,L23

L1 (1,0,0)
L2 (0,1,-1)
L3 (0,1,1)
L4 (0,3,-2)
L5 (0,3,2)
L6 (1,-3,0)
L7 (1,3,0)
L8 (1,2,-2)
L9 (1,-2,2)
L10 (1,2,2)
L11 (1,-2,-2)
L12 (2,9,-6)
L13 (2,-9,6)
L14 (2,9,6)
L15 (2,-9,-6)
L16 (2,5,-2)
L17 (2,-5,-2)
L18 (2,-5,2)
L19 (2,5,2)
L20 (2,-1,-2)
L21 (2,1,-2)
L22 (2,1,2)
L23 (2,-1,2)

EIGHT SIGN ORBITS

i	size	point representative	line representative
1	1	(1,0,0)	(1,0,0)
2	2	(0,1,+/-1)	(0,1,+/-1)
3	2	(0,2,+/-3)	(0,3,+/-2)
4	2	(1,0,+/-1)	(1,+/-3,0)
5	4	(3,2e,2d)	(1,2e,2d)
6	4	(2,2e,3d)	(2,9e,6d)
7	4	(6,2e,5d)	(2,5e,2d)
8	4	(6,2e,d)	(2,e,2d)

Here e,d independently range over {+1,-1}. Total size: 1+2+2+2+4+4+4+4=23.

ORBIT INCIDENCE MATRIX M

	P1	P2	P3	P4	P5	P6	P7	P8
L1	0	2	2	0	0	0	0	0
L2	1	1	0	0	2	0	0	0
L3	1	0	1	0	0	2	0	0
L4	0	0	0	0	0	0	2	2
L5	0	1	0	0	0	1	1	1
L6	0	0	1	0	1	0	1	1
L7	0	0	0	1	1	1	0	1
L8	0	0	0	1	1	1	1	0

Expansion rule. Generate every signed member in the orbit table and declare P incident with L exactly when L dot P = 0. M_{ij} counts the zeros for one representative of line orbit i against point orbit j; every row sums to 4. Weighted column sums are (4,8,8,8,16,16,16,16)=4(1,2,2,2,4,4,4,4), so every point degree is 4.

A line (a,b,c) means ax+by+cz=0; labels agree with coordinates.json.